Preliminary Shielding Assessment - New build

A client-facing preliminary assessment for design development and shielding coordination. It identifies the adopted design basis, preliminary barrier requirements, and the information that must be confirmed before final shielding acceptance.

This report is not a certificate of shielding compliance.

1. Project details

Project name	Example Shielding Assessment
Room name	CT Room 1
Modality	CT
Project type	New build
Project note	Simplified preliminary shielding assessment.
Client	Example Imaging Pty Ltd
Site	123 Example Street, Sydney NSW
Assessor	Health Physics Pty Ltd
Date	2026-06-28
Report status	Draft
Revision	Rev 0

2. Assessment outcome

Overall preliminary outcome	**FAIL**
Barriers assessed	9
Barriers requiring review / redesign	5
Design decision	Do not rely on the current shielding arrangement for construction; redesign the failed barrier(s) and reassess.

3. Design basis

The preliminary result is valid only for the room geometry, source/workload and occupancy assumptions summarised below. These values should be confirmed by the client/design team before detailed design.

3.1 Room and geometry

Room	CT Room 1
Modality	CT
Room dimensions	4.00 m x 6.00 m
FFL to FCL	3.60 m
Source position	2.00 m from west | 4.00 m from south

3.2 Source and workload assumptions

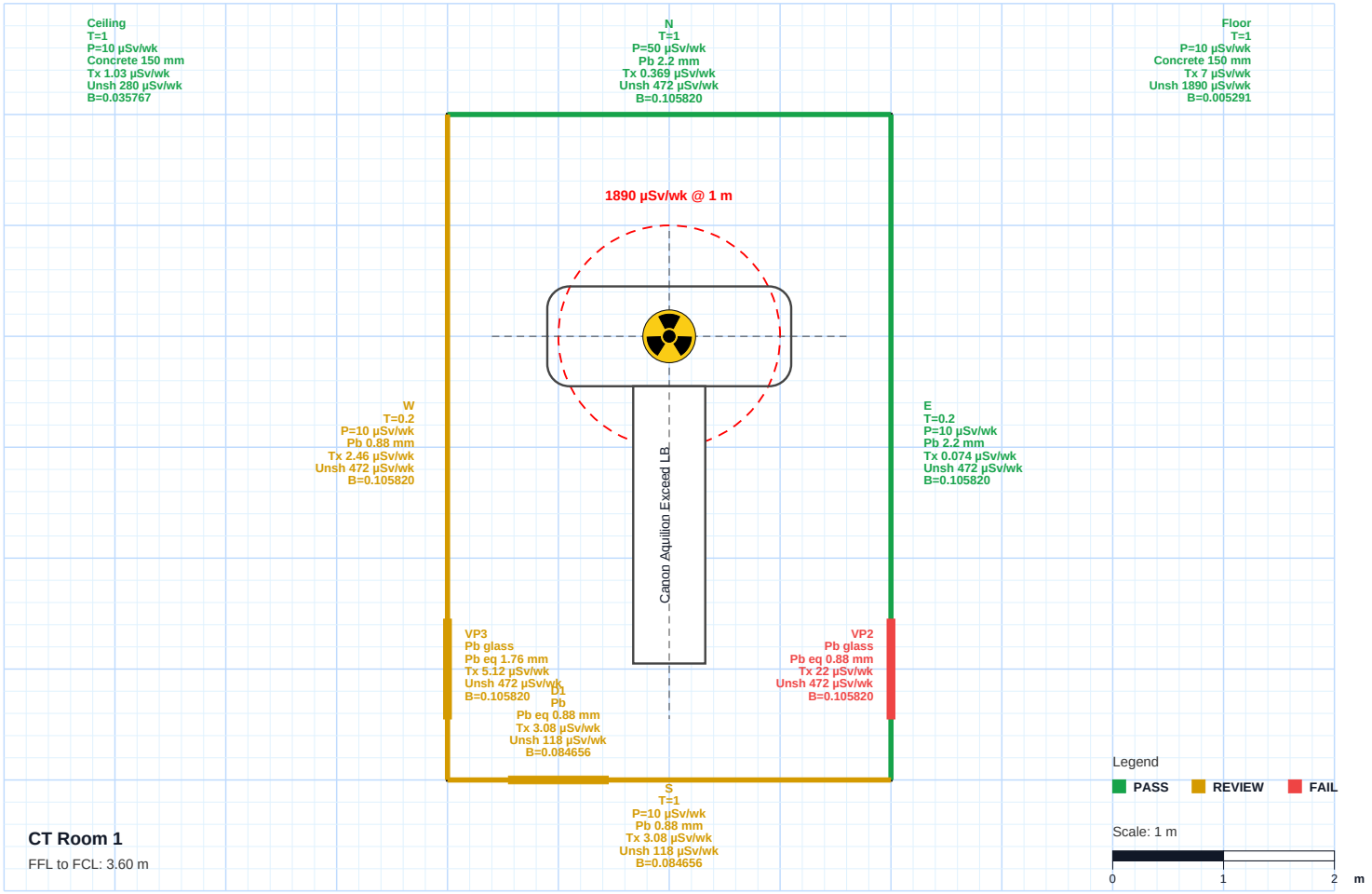
Parameter	Adopted preliminary input
Source	Secondary scatter source diagnostic
Type / radiation	CT scatter | xray 140kvp
Scans per week	60
Workload per scan	630 mAs
ksec at 1 m	0.005 mGy/100 mAs
Tube potential	140 kVp

3.3 Design criteria and occupancy

Barrier	Adjacent area	Distance (m)	Occupancy T	P (uSv/wk)
N wall	Corridor	2.00	1.000	50.0
E wall	Corridor	2.00	0.200	10.0
S wall	Corridor	4.00	1.000	10.0
W wall	Corridor	2.00	0.200	10.0
Floor	Clinical Area	1.00	1.000	10.0

Barrier	Adjacent area	Distance (m)	Occupancy T	P (uSv/wk)
Ceiling	Clinical Area	2.60	1.000	10.0
D1	S wall	4.00	1.000	10.0
VP2	E wall	2.00	0.200	10.0
VP3	W wall	2.00	0.200	10.0

4. Barrier plan



Barrier labels show the entered construction and the calculated preliminary result. The architectural drawing, openings and service penetrations must be coordinated separately before final acceptance.

5. Barrier schedule

The schedule below reports the canonical calculation results for the assessed barriers.

Barrier	Actual (mm)	Role	Distance (m)	T	P (uSv/wk)	Unshielded (uSv/wk)	Transmitted (uSv/wk)	B	Required (mm)	Solver	Status
N wall	2.20	Secondary wall	2.00	1.000	50.000	472.500	0.369	0.105820	0.48	ncrp_ct_secondary_fit	PASS
E wall	2.20	Secondary wall	2.00	0.200	10.000	472.500	0.074	0.105820	0.48	ncrp_ct_secondary_fit	PASS
S wall	0.88	Secondary wall	4.00	1.000	10.000	118.125	3.081	0.084656	0.54	ncrp_ct_secondary_fit	REVIEW
W wall	0.88	Secondary wall	2.00	0.200	10.000	472.500	2.465	0.105820	0.48	ncrp_ct_secondary_fit	REVIEW
Floor	150.00	Floor	1.00	1.000	10.000	1890.000	6.996	0.005291	139.60	ncrp_ct_secondary_fit	PASS
Ceiling	150.00	Ceiling	2.60	1.000	10.000	279.586	1.035	0.035767	84.90	ncrp_ct_secondary_fit	PASS
D1	0.88	Door	4.00	1.000	10.000	118.125	3.081	0.084656	0.54	ncrp_ct_secondary_fit	REVIEW
VP2	0.88	Viewing panel	2.00	0.200	10.000	472.500	22.006	0.105820	1.50	mu_plus_buildup	FAIL
VP3	1.76	Viewing panel	2.00	0.200	10.000	472.500	5.125	0.105820	1.50	mu_plus_buildup	REVIEW

6. Design actions and verification

This page identifies the items that should be closed out before the preliminary assessment is relied upon for construction or final shielding acceptance.

6.1 Design coordination verification

Barrier	Prescribed shielding (mm)	Min. required shielding (mm)	Adjacent area	Occupancy T	Verification status
N wall	2.2 Pb	0.48 Pb	Corridor	1.000	Requires site check
E wall	2.2 Pb	0.48 Pb	Corridor	0.200	Requires site check
S wall	0.88 Pb	0.54 Pb	Corridor	1.000	Requires site check
W wall	0.88 Pb	0.48 Pb	Corridor	0.200	Requires site check
Floor	150 Concrete	139.6 Concrete	Clinical Area	1.000	Requires site check
Ceiling	150 Concrete	84.9 Concrete	Clinical Area	1.000	Requires site check
D1	0.88 Pb	0.54 Pb	S wall	1.000	Requires site check
VP2	0.88 Pb glass	1.5 Pb glass	E wall	0.200	Requires site check
VP3	1.76 Pb glass	1.5 Pb glass	W wall	0.200	Requires site check

6.2 Project coordination actions

Action	Suggested responsibility	Required before
Confirm source workload, occupancy classification and installed barrier thickness.	Client / design team / physicist	Design freeze
Confirm source workload, occupancy classification and installed barrier thickness.	Client / design team / physicist	Design freeze
Confirm source workload, occupancy classification and installed barrier thickness.	Client / design team / physicist	Design freeze
Increase shielding or revise source / occupancy assumptions.	Client / design team / physicist	Design freeze
Confirm source workload, occupancy classification and installed barrier thickness.	Client / design team / physicist	Design freeze
Confirm the current architectural drawing number/revision, room orientation and dimensions.	Architect / client	Detailed assessment
Confirm the use and occupancy of areas beyond each wall, floor and ceiling.	Client / architect	Detailed assessment
Provide door, viewing panel, duct, conduit and service-penetration details so shielding continuity can be assessed.	Architect / builder	Construction documentation
Confirm radionuclide/scanner workload and operating assumptions with the intended operator.	Imaging operator / client	Detailed assessment

6.3 Changes that trigger reassessment

- Any change to scanner model, ksec basis, examination workload or scan technique assumptions.

- Any change to room dimensions, source position, adjacent-area use or occupancy.
- Any change to barrier construction, doors, glazing, penetrations or shielding continuity.
- Addition of radiation-producing equipment or a source not represented in this assessment.

7. Conclusion and next stage

Overall preliminary outcome	**FAIL**
Barrier result	4 PASS | 4 REVIEW | 1 FAIL
Client decision	Do not rely on the current shielding arrangement for construction; redesign the failed barrier(s) and reassess.

7.1 Preliminary conclusion

The preliminary assessment contains 1 barrier failure(s). The affected barriers require redesign and reassessment before the shielding arrangement is accepted for construction.

7.2 Recommended next stage

For a new build, the next stage should coordinate the shielding schedule with architectural drawings, door/window schedules, penetrations and confirmed operating assumptions before issuing a detailed/construction shielding assessment.

7.3 Information normally required for the detailed/final assessment

- Current coordinated architectural drawings showing room dimensions, wall construction, doors, glazing and service penetrations.
- Confirmed use and occupancy of all adjacent areas, including above and below the room.
- Confirmed clinical workload/source assumptions and any additional radiation-producing equipment.

8. Preliminary assessment limitations

- This document is a preliminary shielding assessment for design development and shielding review; it is not a final shielding compliance certificate.
- Final shielding acceptance requires professional review of architectural drawings, construction details, source and workload assumptions, occupancy assumptions, and installed materials.
- Calculated results depend on the entered room geometry, source position, operating assumptions, design goals, occupancy factors, and barrier construction.
- CT results depend on the selected scanner model or verified ksec basis, examination workload, scan technique assumptions, and room geometry.
- Any manual ksec override must be independently verified before final certification.